TEXORA · LECTURE SHEET

Textured Yarn

A flat filament yarn is smooth, dense and cold; texturing permanently crimps, loops or coils its filaments so the yarn gains bulk, cover, warmth and — where required — stretch. This sheet covers the five texturing processes, the three stretch families, and the crimp-contraction test that grades them.

YM-4103 · SPECIAL YARN MANUFACTURING · TOPIC 6 · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

1. Why texture a filament yarn?

Continuous filaments leave the spinneret straight and parallel. A yarn of such filaments is strong and lustrous but has three commercial weaknesses: it packs densely, so fabric cover per gram is poor; its smooth surface feels cold and slippery and traps little air, so insulation and moisture handling are poor; and it has no crimp, so it cannot stretch or recover like a natural-fibre yarn. **Texturing** permanently disrupts the straight filament configuration — by crimping, coiling or looping — to introduce bulk and, in the thermoplastic yarns, controlled stretch. The raw material is usually a thermoplastic filament (polyester, nylon), because heat can set the new configuration permanently; the polyester draw-texturing industry (DTY — draw-textured yarn) is built entirely on this idea, texturing and drawing partially oriented yarn (POY) in one machine pass.

2. False-twist texturing

The dominant process for stretch and bulk in fine apparel yarns. “False” twist means twist is inserted at one point and removed again beyond it, so the delivered yarn has (ideally) zero net twist — yet each filament retains the crimp that the twist-and-heat cycle imposed.

1. **Feed.** Supply yarn (for polyester, POY) is drawn between input and draw rollers, the drawing completing filament orientation.
2. **First heater.** The yarn, twisted as it travels (step 3), passes through a hot zone (commonly around 180–220 °C for polyester, set by polymer and product) where the twisted configuration is heat-set.
3. **Twist unit.** Beyond the heater, **friction discs** (stacks of rotating discs whose edges grip the yarn) spin the yarn at very high speed, inserting twist that runs back through the heater to the feed side. Cooled after the heater — a cooling plate/zone between heater and discs fixes the set — each filament is locked in a helical form.

4. **Untwisting.** Past the discs the inserted twist runs out: the yarn untwists to near zero, but the filaments, set in helices, now coil and crimp against each other — the yarn has bulk and stretch with no residual torque in balanced production.
5. **Second heater (optional).** A second, lower-temperature heater through which the yarn passes under reduced tension/overfeed relaxes part of the crimp: stretch falls, stability rises. Yarns so treated are "set" or stabilised yarns (Section 4).
6. **Delivery.** Intermingling jets may add cohesion points; the yarn is oiled and wound.

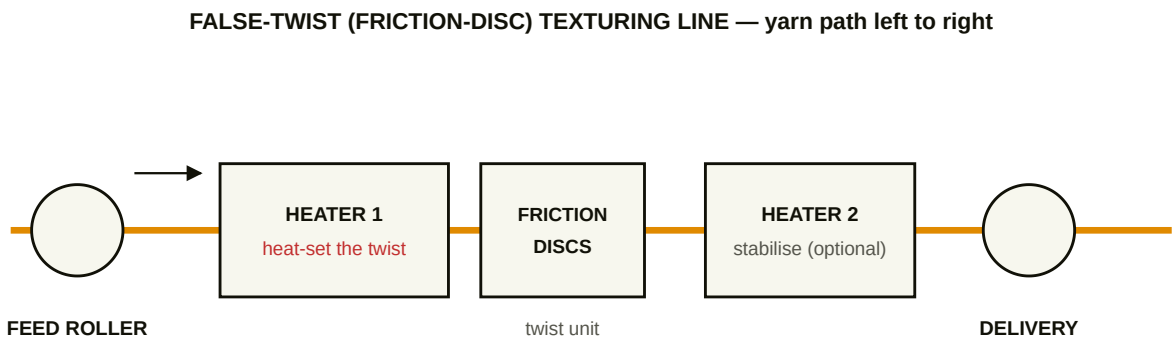


Figure 1 — False-twist texturing line. Feed roller → first heater → cooling → friction-disc twist unit → optional second heater → delivery and winding.

Key settings: D/Y ratio (disc surface speed / yarn speed, controlling twist level), heater temperatures, draw ratio and second-heater overfeed. Fault signatures follow the settings: tight spots and dye streaks from tension or temperature variation, residual torque (spirality in knits) from unbalanced twist, broken filaments ("snow") from excessive disc friction.

3. Other texturing processes

3.1 Air-jet texturing

One or more filament yarns are overfed into a turbulent air jet; the blast opens the filament bundle and blows the surplus length into random loops, which are then intermingled and locked by the converging filaments as the yarn leaves the jet. No heat-setting of twist is involved, so the yarn has **bulk without stretch**, a spun-like, matte surface, and — uniquely — core and effect yarns can be fed together to make combined effects in one pass. Air-jet (taslan-type) yarns serve furnishing, sewing-thread and apparel uses where a natural-fibre look is wanted from filament.

3.2 Stuffer-box crimping

The filament bundle is fed by rollers into a confined box faster than it can leave; the accumulating yarn folds back on itself against a weighted flap, and the sharp folds — heat-set in thermoplastic yarn — become a regular sawtooth (zigzag) crimp. Coarse, bulky crimp suited to carpet and upholstery yarns and tow crimping for staple production.

3.3 Knit-de-knit

The filament yarn is knitted into a small tubular fabric, heat-set in that knitted form, then unravelled. Each stitch position leaves a permanent wavy crimp whose wavelength equals the stitch size — a fine, regular, gentle crimp with moderate bulk and low stretch, historically important for hosiery yarns.

3.4 Edge crimping

The yarn is drawn over a sharp heated edge (the principle of curling ribbon): filaments are deformed asymmetrically — compressed on the edge side, stretched on the outer side — and take a helical coil. Simple and effective for coarse monofilament-type products; largely superseded for apparel multifilaments by false-twist.

PROCESS	DEFORMATION PRINCIPLE	STRETCH?	TYPICAL PRODUCTS
False-twist (friction disc)	Twist + heat-set, then untwist	Yes — adjustable, can be stabilised away	DTY apparel, hosiery, sportswear
Air-jet	Turbulent air loops, intermingled	No — bulk only	Spun-look furnishing, threads
Stuffer-box	Folding in a confined box + set	Low	Carpet, upholstery, tow
Knit-de-knit	Set in knitted loop form, unravelled	Low–moderate	Hosiery, fine wavy yarns
Edge crimping	Asymmetric deformation over an edge	Moderate coil	Coarse / monofilament products

4. Stretch families of textured yarn

- **Stretch yarns** — high crimp, high extension and strong retraction (classic false-twist yarn without stabilisation). For hosiery, swimwear and body-fit garments; in fabric they demand careful handling because the stored stretch distorts cutting and sewing.
- **Modified stretch yarns** — intermediate: part of the crimp is relaxed (e.g. controlled second-heater or steaming treatment) so bulk remains high but extension is moderated and fabric behaviour is

manageable.

- **Stabilised / set yarns (bulk yarns)** — stretch largely removed by a second-heater pass at overfeed; what remains is bulk, cover and softness with near-normal dimensional stability. Most apparel DTY sold for weaving and general knitting is in this family — "textured" in trade usually means bulked and stabilised, not stretchy.

5. Crimp contraction — the grading test

Textured yarn is graded by how much length its crimp stores. In the standard skein test a yarn skein is measured under a high (straightening) load and again under a very light load after a heat/relaxation treatment that lets the crimp develop fully:

$$\text{Crimp contraction \%} = (\text{extended length} - \text{contracted length}) / \text{extended length} \times 100$$

WORKED EXAMPLE — CRIMP CONTRACTION

A skein measures 100.0 cm under the straightening load and 72.0 cm in the fully relaxed, crimp-developed state.
Crimp contraction = (100.0 − 72.0) / 100.0 × 100 = 28.0 / 100.0 × 100 = **28.0%**.
Reading: 28% of the yarn's straight length is stored in crimp — a high-bulk, stretch-family value. A stabilised weaving DTY would show a much lower figure; comparing the two numbers is how a buyer distinguishes a stretch yarn from a set yarn of identical count and filament specification.

Crimp rigidity and crimp stability (how much contraction survives repeated loading, washing and dyeing) complete the test family: a yarn whose crimp washes out was under-set at the first heater, whatever its initial contraction value.

Glossary

Texturing — permanent crimping, coiling or looping of filament yarn for bulk/stretch. **False twist** — twist inserted and removed within one pass, leaving set filament crimp and near-zero net yarn twist. **DTY** — draw-textured yarn: POY drawn and textured in one pass. **D/Y ratio** — friction-disc surface speed divided by yarn speed in false-twist texturing. **Air-jet texturing** — loop formation by turbulent air; bulk without stretch. **Stuffer box** — chamber in which yarn folds and is crimp-set. **Crimp contraction** — percentage length stored in crimp, (extended − contracted) / extended × 100. **Stabilised yarn** — textured yarn whose stretch has been largely set out, retaining bulk.

Key points

• Texturing converts flat filament's weaknesses — low cover, cold handle, no stretch — into bulk, warmth and controlled elasticity.
• False twist: twist is heat-set filament-by-filament, then removed; the crimp, not the twist, is what the customer keeps.
• Air-jet gives bulk with no stretch; stuffer-box gives coarse sawtooth crimp; knit-de-knit copies the stitch as crimp; edge crimping coils by asymmetric deformation.
• Stretch → modified stretch → stabilised is one continuum, travelled by relaxing crimp (typically in a second heater).
• Crimp contraction grades the product: (extended − contracted) / extended × 100; 100 cm → 72 cm = 28%.

Practice

1. A skein extends to 200 mm under load and contracts to 148 mm when relaxed. Calculate its crimp contraction % and state which stretch family it most likely belongs to.
2. Explain, step by step, how a yarn can leave a false-twist machine with almost zero twist yet permanently crimped filaments.
3. A furnishing buyer wants a spun-look, bulky filament yarn with no stretch, to be woven on a high-tension loom. Choose a texturing process and justify it against false-twist.
4. Knitted fabric from a false-twist yarn shows spirality (wale skew). Give the yarn-level cause and one machine-setting change that reduces it.

Answer key

1. Contraction = (200 − 148) / 200 × 100 = 52 / 200 × 100 = **26%**. A contraction this high indicates a **stretch-family** yarn (unstabilised false-twist); a stabilised weaving yarn would contract far less.
2. Twist inserted by the friction discs runs back through the first heater, so filaments are held in helical form while heat sets the polymer in that shape, and cooling fixes it. Beyond the discs the same twist runs out of the yarn, returning net yarn twist to ≈ zero — but each filament keeps its set helical shape, so filaments coil against each other and the yarn shows crimp, bulk and stretch.
3. **Air-jet texturing**. Its loops are locked by intermingling, not by set twist, so the yarn is bulky and spun-like with essentially no elastic stretch to distort under loom tension — whereas false-twist yarn would carry stored stretch (and torque risk) the application does not want.
4. Cause: **residual torque** — unbalanced false twist leaves the yarn lively, and the torque rotates loops in the knit, skewing wales. Reduce it by lowering twist level (lower D/Y ratio), using S- and Z-twist yarns alternately / plied together, or stabilising through the second heater — any one correctly applied earns credit; the direct setting change is a reduced D/Y ratio.